

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		FIRED HEATER DATASHEET API 560 / API 530 / API 535 H-101 - Atmospheric crude charge heater		DOC. No.	CFU-000-ME-DS-003
				REV / DATE	A / 2026-10-02
				STATUS	Issued for review (FEED)
				PREPARED	CLAUDE CODE CHK: - APP: -
				SHEET	1
PROCESS DESIGN DATA					
1	Heater type	2-cell cabin, horizontal tube (twin radiant cells, co-	2	Service	Atmospheric crude charge heater
3	Number required	1	4	Draft	Balanced (FD + ID fans, APH E-120)
5	Total heat absorbed (process)	59.55 MW	6	Steam superheat coil	1.13 MW
7	Total absorbed / fired (LHV)	60.68 / 67.42 MW	8	Efficiency (LHV, guaranteed)	90 %
9	Fluid	Desalted crude	10	Flow rate	569.2 t/h
11	Inlet T / outlet T	274.0 / 365.7 C	12	Inlet P / outlet P	16.99 / 2.37 barg
13	Outlet vaporisation (mass)	53.3 %	14	Allowable / calc. pressure drop	14.6 bar / Vendor
15	Mass velocity (radiant)	1176 kg/m2s	16	Avg. radiant flux / max.	31.5 / 62.4 kW/m2
17	Fouling factor	0.0005 m2K/W	18	Coking allowance / decoking	Steam-air decoking + pigging connections
COMBUSTION DATA					
19	Fuel	Refinery fuel gas, LHV 47 MJ/kg, MW 20, 3.5 barg	20	Fuel consumption	5164 kg/h
21	Excess air (gas)	15 %	22	Flue gas flow	96 t/h
23	Bridgewall temperature (est.)	1044 C	24	Flue gas leaving convection	381 C
25	Flue gas leaving APH / stack	160 C	26	Combustion air preheat (APH E-120)	6.4 MW
27	Ambient design	35 C / -5 C min.	28	Emissions	NOx <= 25 ppmv @ 3 % O2 (TBD by permit)
29	Vac. off-gas burning	-			
COIL DESIGN (API 530)					
30	Section	RADIANT CONVECTION (shock / studded)	31	Tube OD	168.3 mm (6") 168.3 mm
32	Tube wall (min. / selected)	6.12 / 10.97 mm (Sch 80) Sch 40 min.	33	Tube material	A335 P9 (9Cr-1Mo) A335 P5 shock, A106 B studd-
34	Number of tubes	128 64 (2 shock rows)	35	Effective length	18.3 m 18.3 m
36	Passes	8	37	Arrangement	Single row on side walls, 32 per wall per cell 8 p-
38	Tube spacing	337 mm (2 OD) / 1.5 OD from wall	39	Radiant / convection surface	1229 m2 607 m2 bare-equiv.
40	Extended surface	- Studs 12.7 dia x 25 high, 11-13Cr (where TMT > ~	41	Elastic / rupture design pressure	38.5 / 17.0 barg
42	Design metal temperature	510 C (API 530, EOR TMT 494 C + 15 C)	43	Corrosion allowance / design life	3.0 mm / 100,000 h
44	Return bends	Welded 180 deg LR, A234 WP9 (radiant) - plug hea-	45	Tube supports	25Cr-20Ni (A297 HK40) radiant; 50Cr-50Ni-Nb shoc-
46	Hydrotest	57.8 barg (1.5 x design)	47	Coils	SS superheat 2 rows
BURNERS (API 535)					
48	Burner type	Ultra-low-NOx, staged fuel gas, forced draft (prehe-	49	Number / arrangement	16, single row per cell, up-fired from floor (8 per cell)
50	Spacing	2.29 m	51	Heat release design / max	4.21 / 5.27 MW
52	Burner CL to tube CL	2.67 m (API 560 min.)	53	Pilots / ignition	Gas pilots, HEI
54	Flame detection	UV scanner per burner (BMS, NFPA 85 / API 556)	55	Snuffing steam	Firebox + header boxes
MECHANICAL / CASING / STACK					
56	Overall dimensions L x W	21.9 x 12.7 m (plot reservation 26.0 x 12.7 m)	57	Radiant cell (inside) L x W x H	18.9 x 5.85 x 13.0 m
58	Floor elevation / clearance	EL 102.000 / 2.0 m below floor	59	Convection section (inside) W x H	3.09 x 4.2 m
60	Stack ID / top elevation	2.5 m / EL 138.000	61	Refractory - radiant walls	Ceramic fibre modules 200 mm (or IFB/castable 23-
62	Refractory - floor / arch	IFB 65 mm + castable 150 mm / ceramic fibre 250 ~	63	Refractory - convection	Dual-layer castable 150 mm
64	Casing temperature	82 C max at 27 C, 0 m/s wind	65	Structural / casing	CS, ASCE 7 wind 150 mph
66	Platforms	Burner, header-box, convection, stack access	67	APH / fans	E-120 cast-iron/glass tube; K-101A/B FD 90 kW, K--
68	Weight empty / operating	1033 / 1059 t (+/-30 %)			
NOTES					
1. Datasheet per API 560 5th ed. layout. Vendor to complete thermal rating and confirm TMTs.					
2. Tube design: API 530 elastic + rupture (100,000 h) - see CFU-000-ME-CAL-001 S10.					
3. BWT ~1040 C at 0.65 radiant fraction - vendor to optimise radiant surface.					
4. Plot dimensions in equipment.json are inconsistent with 18.3 m tubes for H-201 - see CAL-001 S13.					

GENERIC REFINERY - GULF COAST 100 kBPSP Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU		FIRED HEATER DATASHEET API 560 / API 530 / API 535 H-201 - Vacuum heater		DOC. No.	CFU-000-ME-DS-003
				REV / DATE	A / 2026-10-02
				STATUS	Issued for review (FEED)
				PREPARED	CLAUDE CODE CHK: - APP: -
				SHEET	2
PROCESS DESIGN DATA					
1	Heater type	1-cell cabin, horizontal tube	2	Service	Vacuum heater
3	Number required	1	4	Draft	Natural
5	Total heat absorbed (process)	14.69 MW	6	Steam superheat coil	-
7	Total absorbed / fired (LHV)	14.69 / 16.70 MW	8	Efficiency (LHV, guaranteed)	88 % (see note 3)
9	Fluid	Atmospheric residue + coil steam	10	Flow rate	258.1 t/h
11	Inlet T / outlet T	348.7 / 398.4 C	12	Inlet P / outlet P	18.99 / -0.70 barg
13	Outlet vaporisation (mass)	26.4 %	14	Allowable / calc. pressure drop	19.7 bar / Vendor
15	Mass velocity (radiant)	1066 kg/m2s	16	Avg. radiant flux / max.	25.0 / 49.5 kW/m2
17	Fouling factor	0.0009 m2K/W	18	Coking allowance / decoking	Steam-air decoking + pigging connections
COMBUSTION DATA					
19	Fuel	Refinery fuel gas, LHV 47 MJ/kg, MW 20, 3.5 barg	20	Fuel consumption	1279 kg/h
21	Excess air (gas)	15 %	22	Flue gas flow	24 t/h
23	Bridgewall temperature (est.)	854 C	24	Flue gas leaving convection	399 C
25	Flue gas leaving APH / stack	399 C (see note 3)	26	Combustion air preheat (APH E-120)	-
27	Ambient design	35 C / -5 C min.	28	Emissions	NOx <= 25 ppmv @ 3 % O2 (TBD by permit)
29	Vac. off-gas burning	Dedicated off-gas tips (stream 28, 102 kg/h)			
COIL DESIGN (API 530)					
30	Section	RADIANT CONVECTION (shock / studded)	31	Tube OD	168.3 mm (6") 168.3 mm
32	Tube wall (min. / selected)	6.25 / 10.97 mm (Sch 80) Sch 40 min.	33	Tube material	A335 P9 (9Cr-1Mo) A335 P5 shock, A106 B studd-
34	Number of tubes	40 20 (2 shock rows)	35	Effective length	18.3 m 18.3 m
36	Passes	4	37	Arrangement	Single row on side walls, 20 per wall per cell 4 p-
38	Tube spacing	337 mm (2 OD) / 1.5 OD from wall	39	Radiant / convection surface	382 m2 185 m2 bare-equiv.
40	Extended surface	- Studs 12.7 dia x 25 high, 11-13Cr (where TMT > ~	41	Elastic / rupture design pressure	19.8 / 19.0 barg
42	Design metal temperature	575 C (API 530, EOR TMT 559 C + 15 C)	43	Corrosion allowance / design life	3.0 mm / 100,000 h
44	Return bends	Welded 180 deg LR, A234 WP9 (radiant) - plug hea-	45	Tube supports	25Cr-20Ni (A297 HK40) radiant; 50Cr-50Ni-Nb shoc-
46	Hydrotest	29.7 barg (1.5 x design)	47	Coils	Utility coil 2 rows (1.4 MW, TBD)
BURNERS (API 535)					
48	Burner type	Ultra-low-NOx, staged fuel gas, natural draft	49	Number / arrangement	6, single row per cell, up-fired from floor (6 per cell)
50	Spacing	3.05 m	51	Heat release design / max	2.78 / 3.48 MW
52	Burner CL to tube CL	2.20 m (API 560 min.)	53	Pilots / ignition	Gas pilots, HEI
54	Flame detection	UV scanner per burner (BMS, NFPA 85 / API 556)	55	Snuffing steam	Firebox + header boxes
MECHANICAL / CASING / STACK					
56	Overall dimensions L x W	21.9 x 5.4 m (plot reservation 14.0 x 11.0 m)	57	Radiant cell (inside) L x W x H	18.9 x 4.90 x 8.9 m
58	Floor elevation / clearance	EL 102.000 / 2.0 m below floor	59	Convection section (inside) W x H	1.75 x 3.3 m
60	Stack ID / top elevation	1.2 m / EL 132.000	61	Refractory - radiant walls	Ceramic fibre modules 200 mm (or IFB/castable 23-
62	Refractory - floor / arch	IFB 65 mm + castable 150 mm / ceramic fibre 250 ~	63	Refractory - convection	Dual-layer castable 150 mm
64	Casing temperature	82 C max at 27 C, 0 m/s wind	65	Structural / casing	CS, ASCE 7 wind 150 mph
66	Platforms	Burner, header-box, convection, stack access	67	APH / fans	-
68	Weight empty / operating	328 / 337 t (+/-30 %)			
NOTES					
1. Datasheet per API 560 5th ed. layout. Vendor to complete thermal rating and confirm TMTs.					
2. Tube design: API 530 elastic + rupture (100,000 h) - see CFU-000-ME-CAL-001 S10.					
3. 88 % LHV needs ~1.4 MW utility coil or APH (not in equipment list) - HOLD.					
4. Plot dimensions in equipment.json are inconsistent with 18.3 m tubes for H-201 - see CAL-001 S13.					